Clinical Risk Assessment Report: Patient 31

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 29.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.34x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the 75th Percentile Tumor Density (CONVENTIONAL_HUQ3), which elevated the patient's absolute risk by +2.7%. Biologically, this feature reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

2. Key Pathological & Imaging Drivers (Elevating Risk)

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.67 [Avg (67th %ile)] (+2.7%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 1.67 [Top 2%] (+2.5%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.42 [Top 5%] (+1.9%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.72 [Top 23%] (+1.9%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.2 [Bottom 21%] (+1.8%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.55 [Top 15%] (+1.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.76 [Avg (74th %ile)] (+1.2%)**

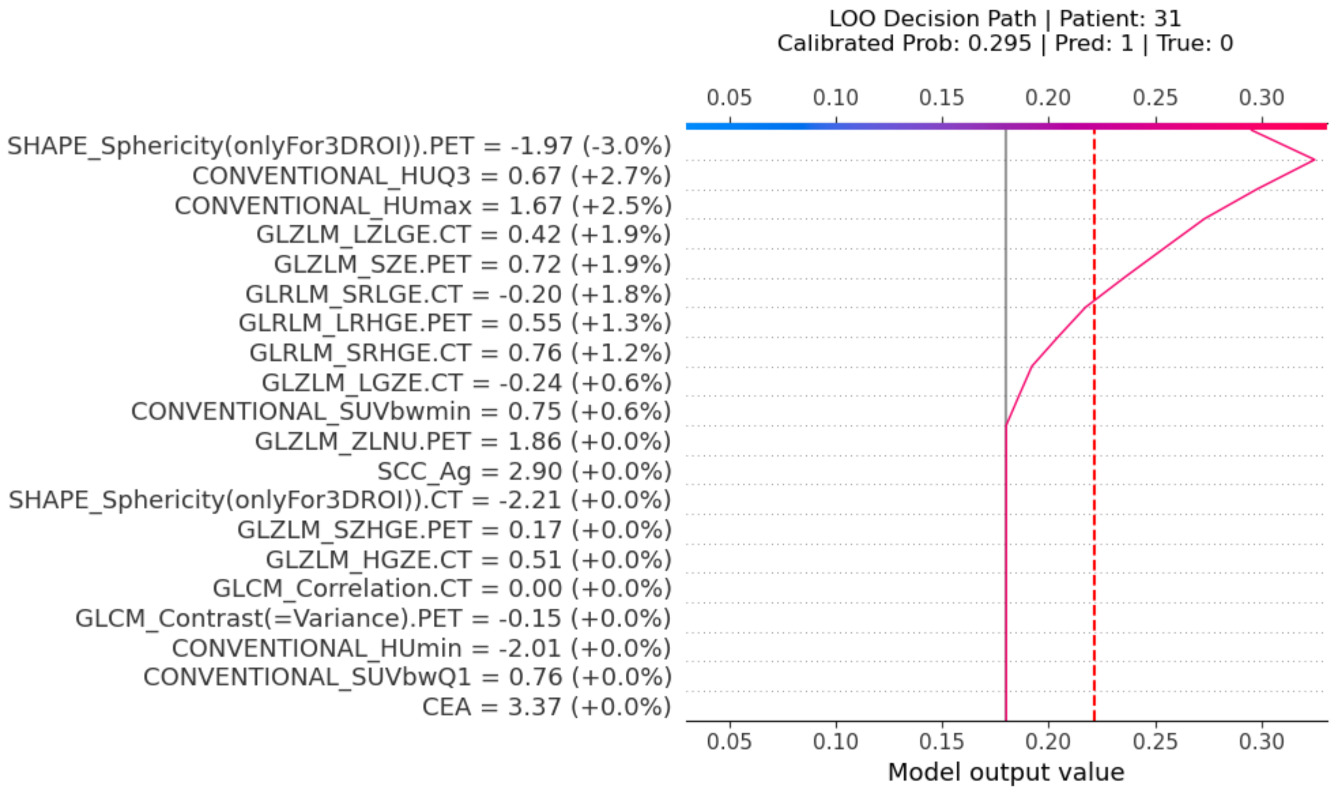
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: -1.97 [Bottom 6%] (-3.0%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.